

Pythagoras' Theorem

WorksheetHub AI
Mathematics · Grade 7

Unit 3 · Geometry

Hard

Pythagoras theorem

Name: _____

Date: _____

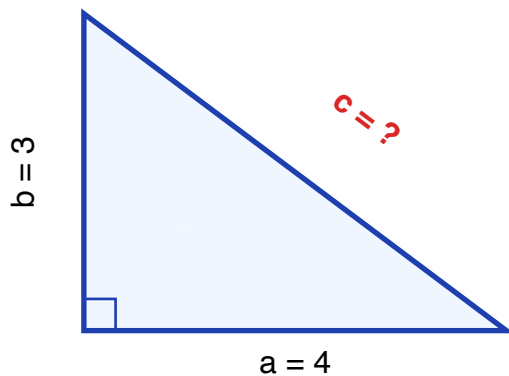
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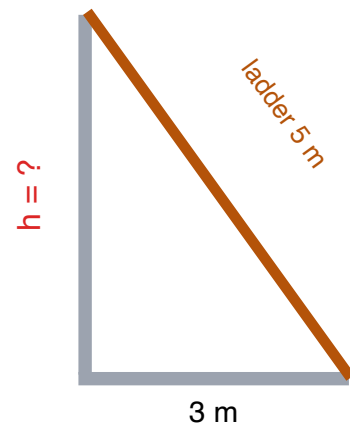
Remember

In a right-angled triangle: $a^2 + b^2 = c^2$ where c is the hypotenuse.

To find c : add the squares, then square root. To find a short side: **subtract** the squares, then square root.



c is the HYPOTENUSE: the longest side, opposite the right angle



Real-life right-angled triangle



Worked example

Find c : $c^2 = 3^2 + 4^2 = 9 + 16 = 25 \rightarrow c = \sqrt{25} = 5$



Famous Pythagorean triples

3-4-5 | 5-12-13 | 8-15-17 | 7-24-25 (and their multiples, like 6-8-10)

A. Find the hypotenuse

- 1 Legs 6 cm and 8 cm. Find the hypotenuse.
Answer:
- 2 Legs 5 cm and 12 cm. Find the hypotenuse.
Answer:
- 3 Legs 6 cm and 7 cm. Find the hypotenuse to 1 decimal place.
Answer:

B. Find a shorter side

- 4 Hypotenuse 13 cm, one leg 5 cm. Find the other leg.
Answer:
- 5 Hypotenuse 10 cm, one leg 6 cm. Find the other leg.
Answer:
- 6 Hypotenuse 9 cm, one leg 4 cm. Find the other leg to 1 d.p.
Answer:

C. Is it a right-angled triangle?

- 7 Sides 7, 24, 25.
Answer:
- 8 Sides 6, 8, 11.
Answer:

D. Real-life problems

- 9 A 5 m ladder leans against a wall with its foot 3 m from the wall (see picture). How high up the wall does it reach?
Answer:
- 10 A rectangular TV screen is 24 cm wide and 7 cm high. Find the length of its diagonal.
Answer:
- 11 Mia walks 80 m north then 60 m east. How far is she from where she started in a straight line?
Answer:

★ Challenge

- 12 Find the distance between the points (1, 2) and (4, 6). (Hint: draw a right-angled triangle.)
Answer:
- 13 A square has a diagonal of 10 cm. Find the side length to 1 d.p.
Answer:

